

Artificial Intelligence Lab

Project Presentation

“Avoid-3: Adversarial Search in a Misere Context”

Engineering depth-limited Minimax and Alpha Beta Pruning for a 4x4 state-space explosion.

Presented by

Student Name	Student ID
Helal Uddin Patwary	0432410005101086
Gaus Saraf Murady	0432410005101088
Prosadi Zaman	0432410005101109
Habiba Islam	0432410005101125

Presented to

Faculty Name	Designation
Md. Ismail	Lecturer
Md. Faysal	Lecturer

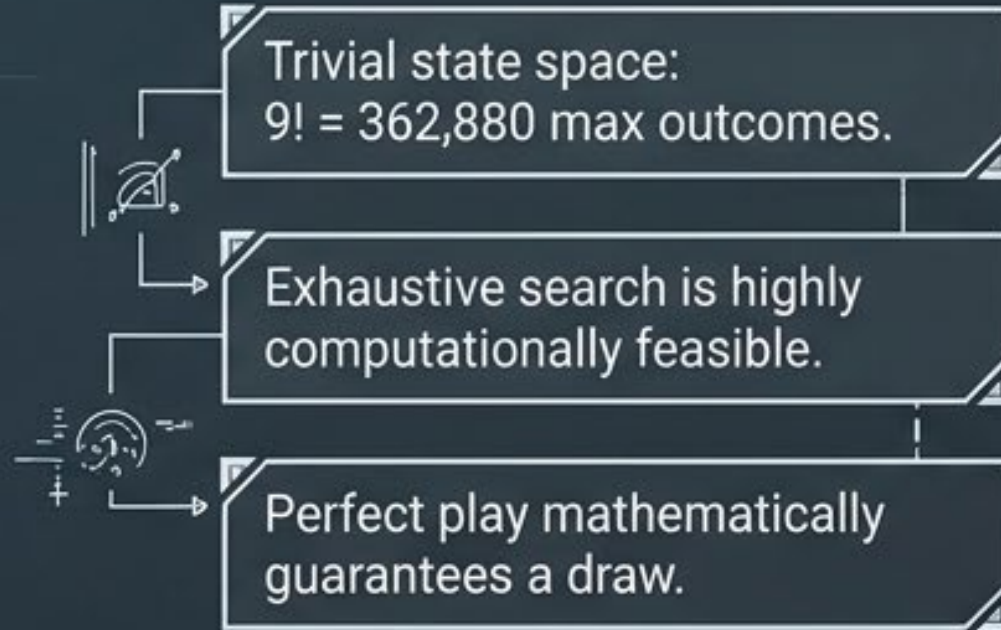
CSE-55, Spring-2026

Course Code: CSE0613312

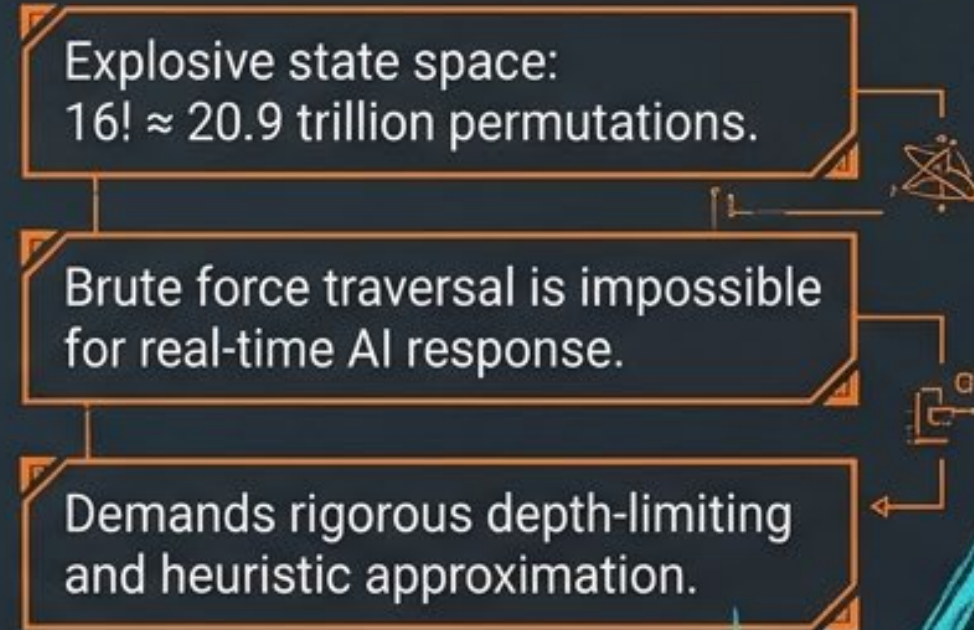
Submission: 19/05/26

The 16! State Space Explosion

Classic 3x3 Tic-Tac-Toe



Avoid-3 4x4 Variant



$O(b^d)$

The diagram shows a large, complex game tree for the Avoid-3 4x4 variant. It starts with a root node, branches into 16 nodes, then into 15 nodes, and finally into 14 nodes. The tree is extremely dense and complex, illustrating an explosive state space. A large arrow points to the tree, and the text $O(b^d)$ is written next to it, indicating the complexity of the state space.

Inverting the Objective: The Misère Ruleset



Rule 1: The Instant Loss

Forming ANY 3 consecutive tiles (row, column, or diagonal) results in immediate elimination.

Rule 2: The Board-Full Win

If players survive until all 16 cells are filled, the winner is determined by the highest count of unmatched tiles (tiles not part of any 3-in-a-row).

The AI Mandate

Minimax must be fundamentally adapted to actively avoid terminal traps while maximizing spatial distribution for the tiebreaker.

Architecture: Pure Python Foundation

Module 1:

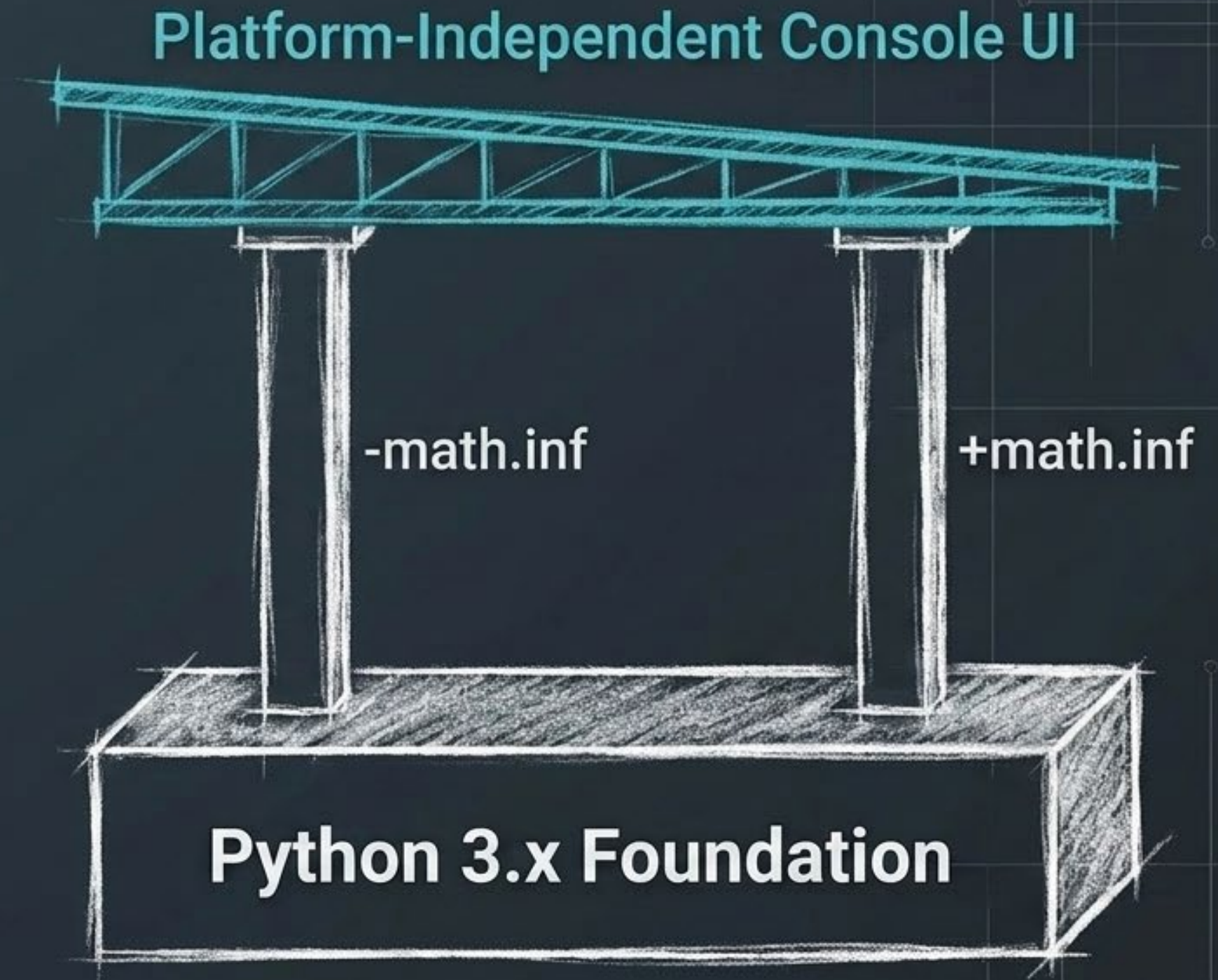
Foundation layer: Built entirely in Python 3.x with zero external dependencies.

Module 2:

Core Logic Support: Utilizing the standard library `math.inf` for IEEE 754 positive/negative infinity, critical for optimal initialization of Minimax maximization loops.

Module 3: User Interface:

A lightweight, platform-independent console environment handling input validation and dynamic 1-based to 0-based index translation.



Board Representation & Loss Line Enumeration

Memory Efficiency

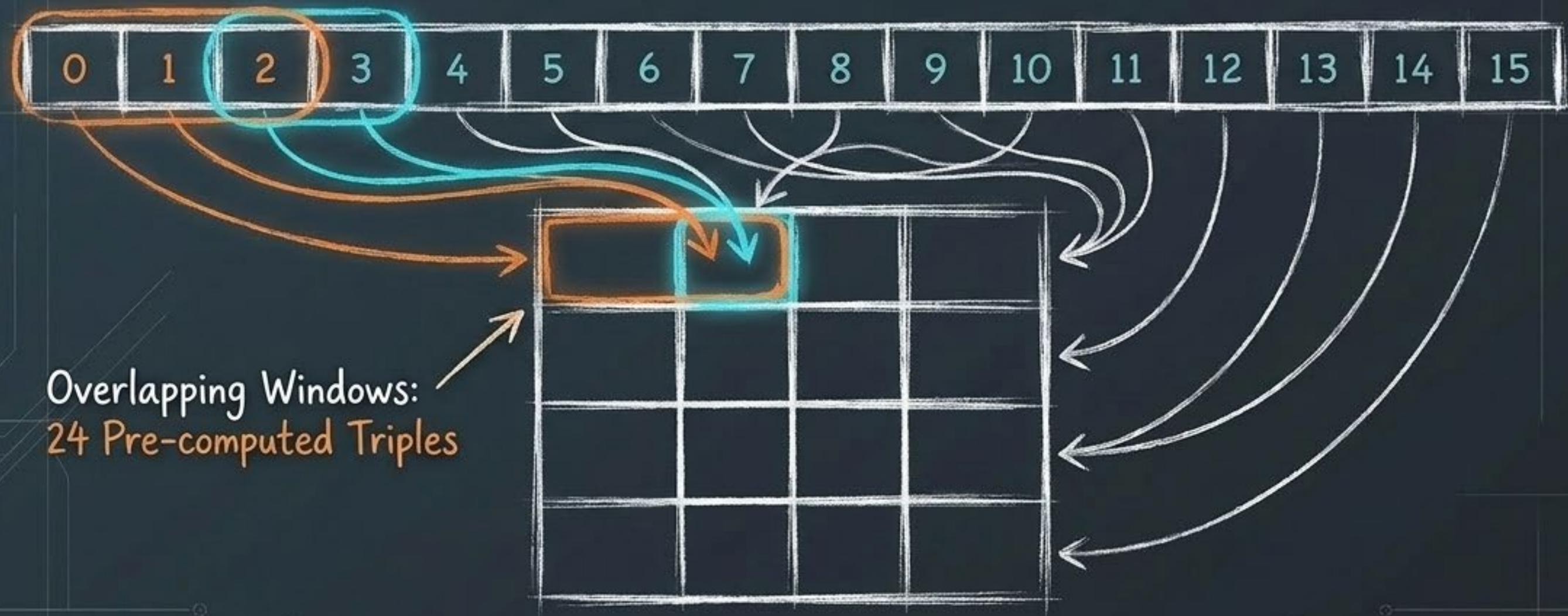
The 4x4 matrix is flattened into a single, contiguous 1D array (`[' '] * 16`). Row/column positions map directly to $r \times 4 + c$.

The Overlap Challenge

A 4-cell dimension creates overlapping 3-cell windows within the same row or column.

Enumerating LOSS_LINES

The engine pre-computes 24 distinct failure triples spanning rows, columns, and both diagonal directions.



The AI Core: Depth-Limited Minimax

The Engine

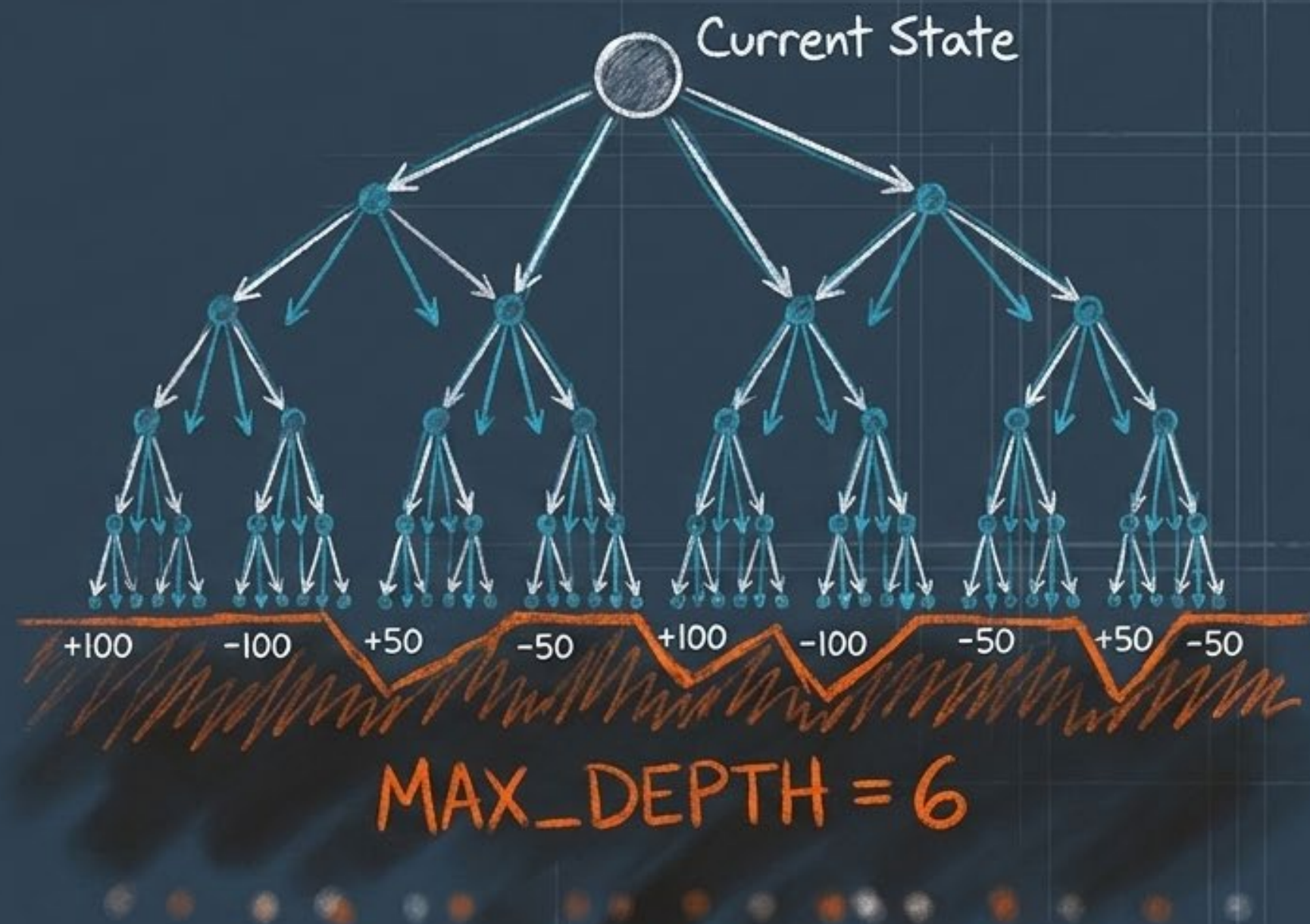
A recursive adversarial search where AI ('O') acts as the Maximizer and Human ('X') acts as the Minimizer.

The Depth Constraint

Hard-capped at 6 plies. The AI computes exactly 6 future moves before deferring to heuristic evaluation, ensuring real-time console response without crashing the system.

Terminal Scoring Dynamics

True terminal states are scored at +/- 100 or 50. Crucially, the current depth is subtracted from the score to enforce faster wins and slower losses.



Optimization: Alpha-Beta Pruning

The Mechanism

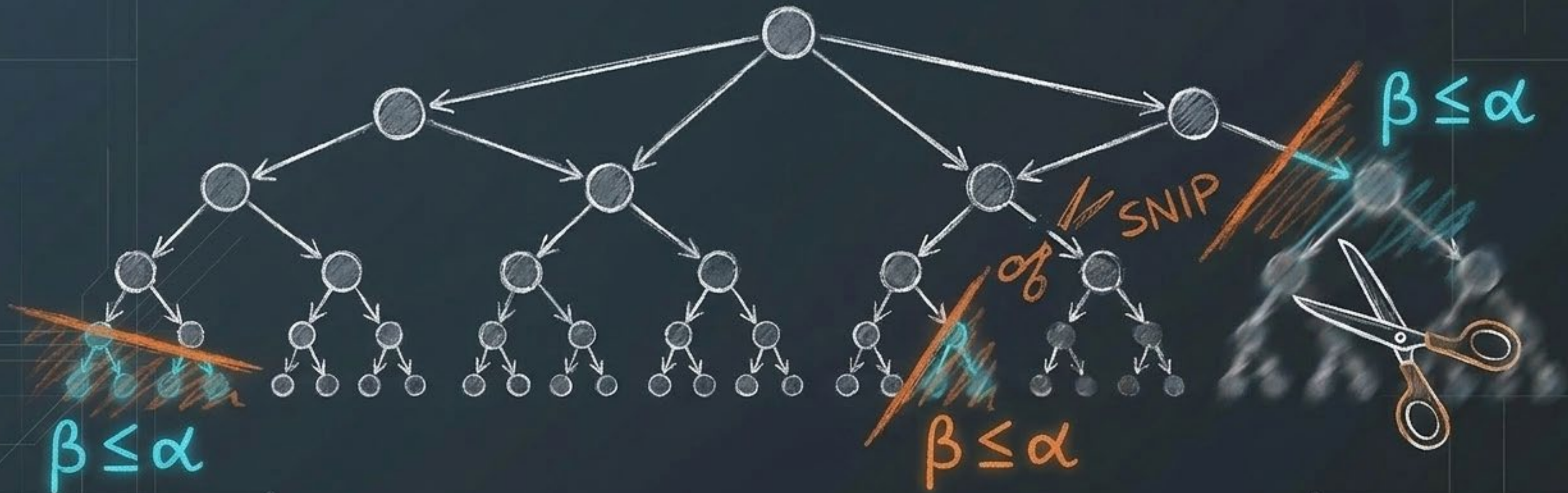
Symmetric bounds tracking.
 α (Maximizer's best guarantee) vs
 β (Minimizer's best guarantee).

The Cut-off Logic

When $\beta \leq \alpha$ at any node, it mathematically proves the opponent will never allow this branch to be reached. Sibling nodes are aggressively pruned.

The Result

Reduces effective branching factor from b to $\sqrt{b}$. This optimization is what strictly enables the engine to reach a 6-ply depth within the standard compute budget.



Heuristic Evaluation: Scoring the Fog of War

The Mathematical Instinct

At **ply 6**, non-terminal boards rely on a bespoke approximation formula to judge spatial dominance vs immediate threats.

$$(\text{ai_unmatched} - \text{human_unmatched}) + (\text{human_danger} - \text{ai_danger}) * 0.5$$

- ✂ **Unmatched Advantage:** Prioritizes long-term spatial dominance for the board-fill tiebreaker.
- ✂ **Danger Penalties:** Sub-dominant weight (0.5) prevents the AI from becoming overly defensive and sacrificing board coverage merely to dodge theoretical, non-immediate threats.



System Boundaries & Current Limitations

The Horizon Problem

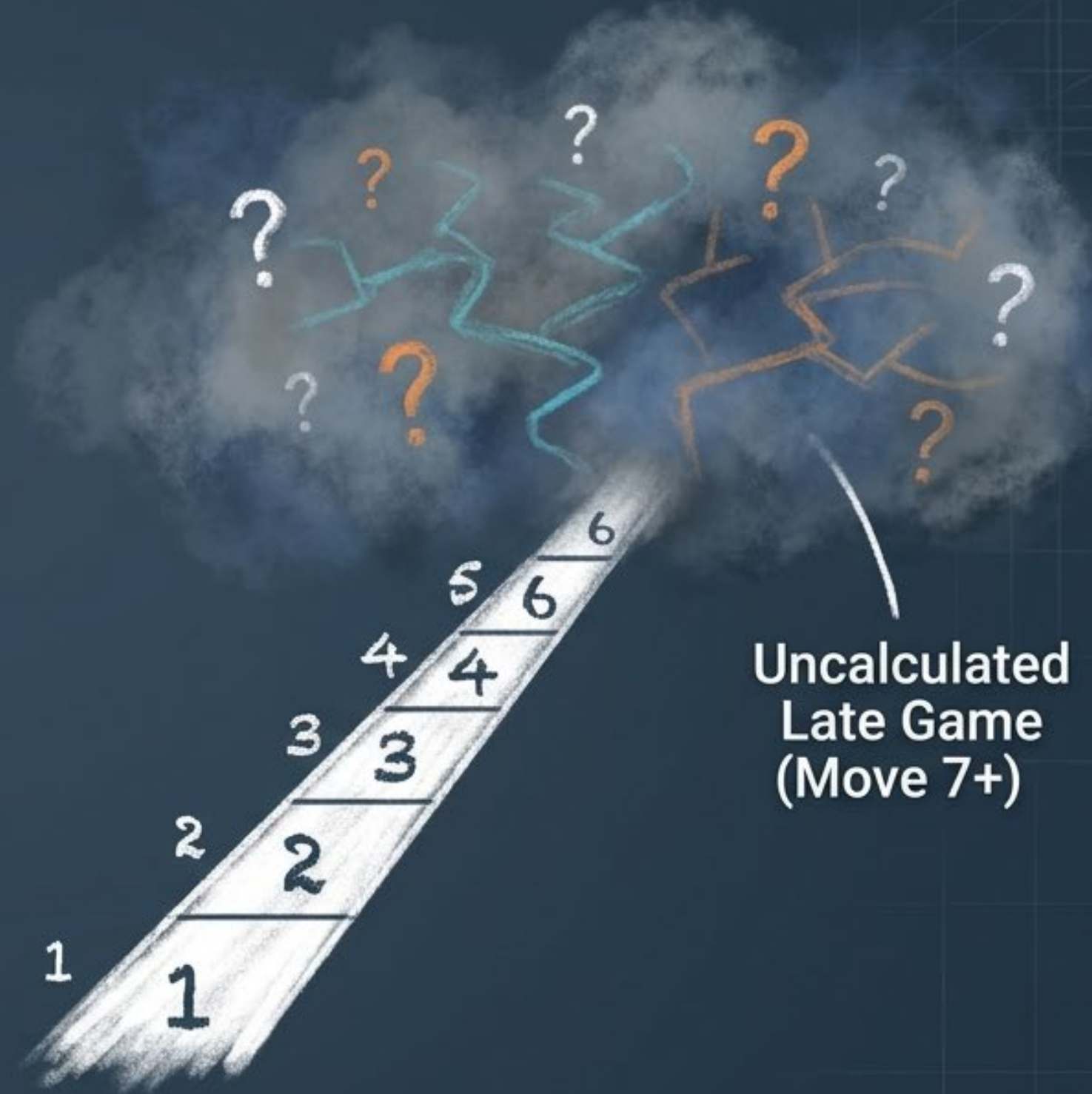
At $\text{MAX_DEPTH} = 6$, the AI cannot see the full 16-move sequence from an empty board. Early-to-mid game decisions are purely heuristic approximations.

Heuristic Blind Spots

The current danger formula evaluates player threats independently. It does not actively model complex interplay, such as deliberately forcing the opponent to complete a sequence (blocking strategies).

Cold Starts

Lack of a pre-computed opening book means the first move is calculated purely on the 6-ply horizon, yielding openings that may not be theoretically optimal.



Evolution Path: Further Work



Iterative Deepening

Replacing the hard depth limit with time-constrained iterative searches, allowing the AI to look deeper when the board simplifies and branching factors shrink.

Neural Heuristic Training

Gathering self-play data to train a neural network, replacing the hardcoded 0.5 danger weight with learned, dynamic spatial evaluations.

GUI Development

Migrating from console constraints to a full graphical user interface to visualize the state space and broaden user accessibility.

Thank You